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Student's Weekly Reportof Internship

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Enrollment Number:	201260107012			
Name Of Organization:	State Cyber Cell-CID			
External Guide Name:	Mr. Nikhil Prajapati			
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Work done in last Week with description :

Week 4 (Date : 23th February 2024 to 29th February 2024) :

1) Digital forensic department tools:- UFAD , OXYGEN

Plans for next week:

1) FALCON TOOL , FTK TOOL

Task

- **Digital forensic department**

- State cyber cells typically utilize a variety of digital forensic tools to investigate cybercrimes, collect evidence, and analyze digital artifacts. Here's a list of commonly used digital forensic tools by state cyber cells:

  - EnCase Forensic: EnCase Forensic is a widely used digital forensic tool for acquiring, analyzing, and reporting on digital evidence from various sources, including computers, mobile devices, and external storage media.
  - FTK (Forensic Toolkit): FTK is a comprehensive digital forensic tool that enables forensic examiners to conduct forensic analysis, data recovery, and evidence examination on computers and mobile devices.
  - Autopsy: Autopsy is an open-source digital forensic platform used for analyzing disk images and other digital artifacts to recover evidence, investigate security incidents, and perform file system analysis.
  - X-Ways Forensics: X-Ways Forensics is a powerful forensic tool that offers advanced features for disk imaging, data carving, metadata analysis, and timeline reconstruction to support forensic investigations.
  - Sleuth Kit (TSK): The Sleuth Kit is an open-source digital forensic toolkit that provides command-line tools and libraries for examining disk images, file systems, and data structures to recover evidence and perform forensic analysis.
  - Cellebrite UFED (Universal Forensic Extraction Device): Cellebrite UFED is a mobile forensic tool used for extracting data from mobile devices, including smartphones, tablets, and GPS devices, to recover evidence and analyze digital artifacts.
  - Oxygen Forensic Detective: Oxygen Forensic Detective is a comprehensive forensic tool designed for extracting, analyzing, and reporting on digital evidence from mobile devices, cloud services, and IoT (Internet of Things) devices.
  - AccessData Forensic Toolkit (ADFT): AccessData Forensic Toolkit is a digital forensic tool used for acquiring, analyzing, and processing digital evidence from computers, mobile devices, and network storage to support forensic investigations.
  - Magnet AXIOM: Magnet AXIOM is a digital forensic platform that offers integrated capabilities for acquiring, analyzing, and reporting on digital evidence from computers, mobile devices, and cloud services to support investigations.
  - Volatility: Volatility is an open-source memory forensics framework used for analyzing volatile memory (RAM) to identify and extract artifacts related to running processes, network connections, and malware activity during forensic investigations.
  - Bulk Extractor: Bulk Extractor is a digital forensic tool used for scanning disk images and memory dumps to extract information such as email addresses, credit card numbers, and other artifacts that may be relevant to forensic investigations.

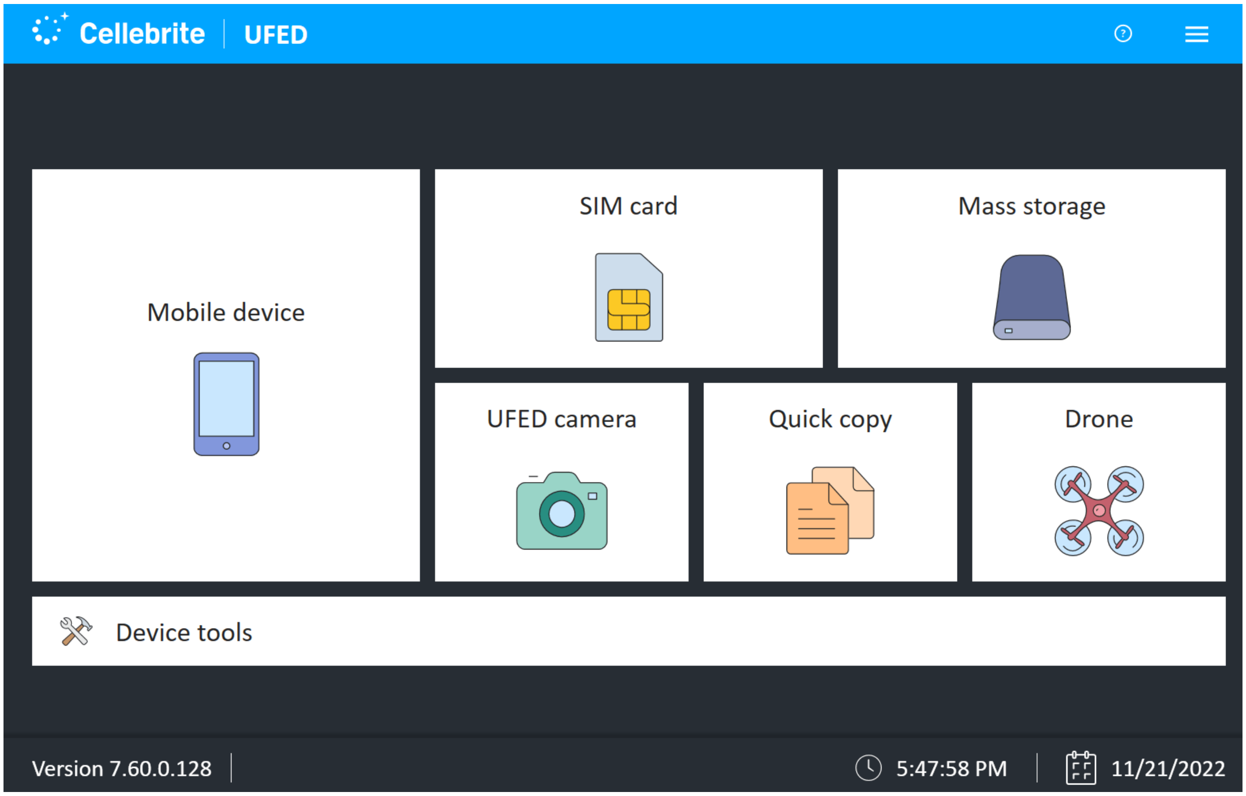
- Registry Viewer (Registry Recon): Registry Viewer is a digital forensic tool used for analyzing Windows registry hives to recover evidence, identify user activity, and detect malware infections during forensic examinations.

- **UFED TOOL**

- UFED (Universal Forensic Extraction Device) is a digital forensic tool developed by Cellebrite, a leading provider of digital intelligence solutions. UFED is widely used by law enforcement agencies, government organizations, and cybersecurity professionals for mobile device forensics and data extraction.

- UFED enables investigators to perform comprehensive forensic examinations on a wide range of mobile devices, including smartphones, feature phones, tablets, and GPS devices. It provides advanced capabilities for acquiring, analyzing, and reporting on digital evidence stored on mobile devices, including:

- Data Extraction: UFED can extract a wide range of data types from mobile devices, including call logs, text messages, emails, contacts, photos, videos, social media messages, application data, GPS locations, and internet browsing history.
- Physical Extraction: UFED can perform physical extractions of data from mobile devices, allowing investigators to access and recover deleted data, hidden files, and system files that may not be accessible through logical extraction methods.
- File System Analysis: UFED provides tools for analyzing file systems and data structures on mobile devices, enabling forensic examiners to reconstruct file paths, analyze metadata, and identify evidence of tampering or manipulation.
- Password and PIN Bypass: UFED can bypass password and PIN locks on mobile devices to access encrypted data and perform forensic examinations, even on devices protected with strong encryption mechanisms.
- Cloud Extraction: UFED supports cloud extraction capabilities, allowing investigators to access and retrieve data stored in cloud services such as iCloud, Google Drive, Dropbox, and Microsoft OneDrive associated with the mobile device.
- Reporting and Documentation: UFED generates comprehensive reports and documentation of forensic examinations, including detailed summaries of extracted data, analysis findings, and evidentiary artifacts, which can be used in legal proceedings.
- UFED is equipped with intuitive user interfaces and workflows designed to streamline the forensic examination process and facilitate collaboration among forensic examiners and investigators. It supports a wide range of mobile device models, operating systems, and encryption methods, ensuring compatibility with the latest technologies and devices.



➤ **OXYGEN**

Oxygen Forensic Detective is a comprehensive digital forensic tool developed by Oxygen Forensics, Inc. It is widely used by law enforcement agencies, government organizations, and digital forensic professionals worldwide for extracting, analyzing, and reporting on digital evidence from various sources, including mobile devices, cloud services, and IoT (Internet of Things) devices.

Here are some key features and capabilities of Oxygen Forensic Detective:

Mobile Device Forensics: Oxygen Forensic Detective supports the extraction and analysis of data from a wide range of mobile devices, including smartphones, feature phones, tablets, and wearables. It can extract data from devices running on iOS, Android, Windows Phone, BlackBerry, and other mobile operating systems.

Data Extraction: Oxygen Forensic Detective can extract a wealth of data from mobile devices, including call logs, text messages, emails, contacts, photos, videos, social media messages, internet browsing history, GPS locations, application data, and more.

Physical and Logical Acquisition: The tool offers both physical and logical acquisition methods for extracting data from mobile devices. Physical acquisition allows forensic examiners to access and recover deleted data, hidden files, and system artifacts that may not be accessible through logical extraction methods.

Cloud Forensics: Oxygen Forensic Detective supports cloud forensics capabilities, enabling investigators to extract and analyze data stored in cloud services such as iCloud, Google Drive, Dropbox, Microsoft OneDrive, and social media platforms associated with the mobile device.

IoT Device Forensics: The tool provides support for analyzing data from IoT devices, including smartwatches, fitness trackers, smart home devices, and automotive systems. It can extract data from IoT devices connected to the mobile device, such as fitness activity logs, location history, and device settings.

Data Analysis and Visualization: Oxygen Forensic Detective offers powerful data analysis and visualization tools that allow forensic examiners to search, filter, and analyze extracted data efficiently. It provides graphical timelines, charts, maps, and other visualization tools to help investigators understand relationships and patterns in the data.

Reporting and Documentation: The tool generates comprehensive reports and documentation of forensic examinations, including detailed summaries of extracted data, analysis findings, and evidentiary artifacts. Reports can be customized and exported in various formats for use in legal proceedings.

Total Working Hrs: ~~25 Hours~~ **Signature of Student:**

The above entries are correct and the grading of work done by Trainee is

Excellent	Very Good	Good	Fair	Below Average	Poor

Signature of External Guide with Company Seal: **Signature of Internal Guide**

Date: **Date:**